

18-661: Introduction to ML for Engineers

MLE/MAP and Linear Algebra Review

Spring 2026

ECE – Carnegie Mellon University

Announcements

- No class on Monday Jan 19th, MLK day
- Self-assessment math quiz available on Gradescope until Friday. Recitations on Friday will go over the math quiz questions and solutions.
- ↳ • Math review resources posted on Piazza
- • HW1 will be released later this week

Registration

- If you're not registered, we encourage you to stay patient (waitlists are gradually clearing).
- You are welcome to keep attending the lectures until the waitlists are sorted out

**Direct all waitlist-related questions to:
ece-waitlists@andrew.cmu.edu**

1. Recap: Probability Review
- 2. A Simple Learning Problem: MLE/MAP Estimation
3. Linear Algebra Review

Recap: Probability Review

Probability Terminology

Name	Type	Symbol	Meaning
→ Sample Space	set	Ω, S	possible outcomes
→ Event Space	a set of subsets (of Ω)	$\mathcal{F}, E$	the events that have probabilities
→ Probability Measure	measure	P, π	assigns <u>probabilities</u> to events
<u>Probability Space</u>	a triple	$(\Omega, \mathcal{F}, P)$	

- Rolling a fair die

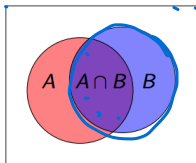
- $\Omega: \{1, 2, 3, 4, 5, 6\}$
- • $\mathcal{F} = \{\{1\}, \{2\}, \dots, \{1, 2\}, \dots, \{1, 2, 3\}, \dots, \{1, 2, 3, 4, 5, 6\}, \{\}\}$
- $P(\text{rolling an odd number}) = P(\{1, 3, 5\}) = \frac{1}{2}$

- Tossing a fair coin twice

- $\Omega: \{HH, HT, TH, TT\}$
- $\mathcal{F} = \{\{HH\}, \{HT\}, \dots, \{HH, HT\}, \dots, \{HH, HT, TH, TT\}, \{\}\}$
- $P(\text{first flip is heads}) = P(\{HH, HT\}) = \frac{1}{2}$

Axioms of Probability

The probability measure satisfies the following properties:



- $0 \leq P(A) \leq 1$
- $P(\Omega) = 1, P(\emptyset) = 0$
- $P(\underline{A \cup B}) = \underline{P(A)} + \underline{P(B)} - \underline{P(A \cap B)}$

Consider two events $A, B \in \mathcal{F}$.

- The **conditional probability** of A given B is given by:
 $\underline{P(A|B)} = \frac{P(A \cap B)}{P(B)} \quad \hookrightarrow \quad P(B|A) = \frac{P(A \cap B)}{P(A)}$
- **Bayes rule:** $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$, which can be derived from
 $\underline{P(B|A)P(A) = P(A \cap B) = P(A|B)P(B)}$

Random Variables

Formally, a random variable is a function $X : \underline{\Omega} \rightarrow \mathbb{R}$ that assigns a numerical value to each outcome s within a probability space $(\Omega, \mathcal{F}, P)$.

Example: Rolling a fair die

- $\Omega: \{1, 2, 3, 4, 5, 6\}$
- $\mathcal{F} = \{\{1\}, \{2\}, \dots, \{1, 2\}, \dots, \{1, 2, 3\}, \dots, \{1, 2, 3, 4, 5, 6\}, \{\}\}$
- $X(S)$ = S for each $S \in \Omega$

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- $X(S) = S$ for each $S \in \Omega$

The **expectation** of a random variable, $\mathbb{E}[X]$, is defined as $\sum_{s \in \Omega} X(s)P(s)$, i.e., the average value of X .

- $\mathbb{E}[X] = \sum_{s=1}^6 sP(s) = \frac{1+2+3+4+5+6}{6} = \underline{\underline{\frac{7}{2}}}$

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The **variance** of a random variable is $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$, which measures how much X can vary from its expected value $\mathbb{E}[X]$.

- $\text{Var}[X] = \sum_{s=1}^6 \frac{1}{6}(s - \frac{7}{2})^2 = \frac{35}{12}$

Exercise
for a fair coin
tossed once
 $X = \begin{cases} 0 & \text{tails} \\ 1 & \text{heads} \end{cases}$

$\mathbb{E}(X) = 1/2$

$\text{Var}(X) = ?$
 $(0 - \frac{1}{2})^2 \frac{1}{2} + (1 - \frac{1}{2})^2 \frac{1}{2} = \frac{1}{4}$

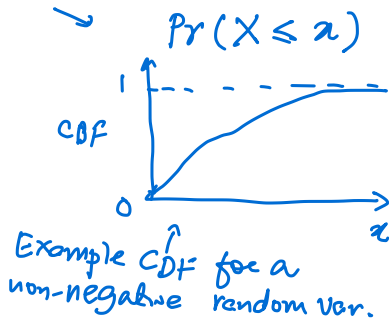
Some Other Concepts that You Should Know

Bernoulli
Poisson
Geometric
Binomial

Gaussian, Exponential

- Discrete and continuous random variables
- PMF (probability mass function), PDF (probability density function), CDF (cumulative distribution function) of random variables
- Entropy of a random variable

Some of these will be covered in HW1



A Simple Learning Problem: MLE/MAP Estimation

- Scenario: You find a coin on the ground.



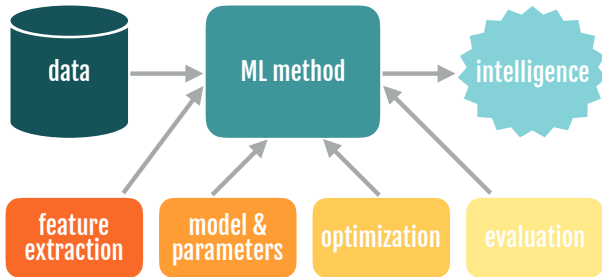
- *You ask yourself: Is this a fair or biased coin? What is the probability that I will flip a heads?*

- You flip the coin 10 times ...

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- It comes up as 'H' 8 times and 'T' 2 times

- You flip the coin 10 times ...
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- Can we learn the bias of the coin from this data?

Recall: Machine Learning Pipeline



Two methods that we will discuss today:

- Maximum likelihood Estimation (MLE)
- Maximum a posteriori Estimation (MAP)

Maximum Likelihood Estimation (MLE)

- **Data:** Observed sequence D of n_H heads and n_T tails

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- **Model:** Each flip follows a Bernoulli distribution

$$\underline{P(H)} = \underline{\theta}, \underline{P(T)} = 1 - \theta, \theta \in [0, 1]$$

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Thus, the likelihood of observing sequence D is

H H T H H T H H H H

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Thus, the likelihood of observing sequence D is

$$P(D | \theta) = \theta^{n_H} (1 - \theta)^{n_T}$$

Because coin tosses
are independent
we can write this
as a product of
 θ 's & $(1-\theta)$'s

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Thus, the likelihood of observing sequence D is

$$P(D \mid \theta) = \theta^{n_H}(1 - \theta)^{n_T}$$

- **Question:** Given this model and the data we've observed, can we calculate an estimate of θ ?

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Thus, the likelihood of observing sequence D is

$$P(D \mid \theta) = \theta^{n_H} (1 - \theta)^{n_T}$$

- **Question:** Given this model and the data we've observed, can we calculate an estimate of θ ?
- **MLE:** Choose θ that maximizes the *likelihood* of the observed data

$$\hat{\theta}_{MLE} = \arg \max_{\theta} P(D \mid \theta)$$

How to Solve?

- $\log(x)$ is a monotone increasing function; will not affect the $\arg \max$

$$\begin{aligned}\hat{\theta}_{MLE} &= \arg \max_{\theta} \underline{P(D | \theta)} \\ &= \arg \max_{\theta} \log \underline{P(D | \theta)} \\ &= \arg \max_{\theta} \log (\theta^{n_H} (1 - \theta)^{n_T})\end{aligned}$$

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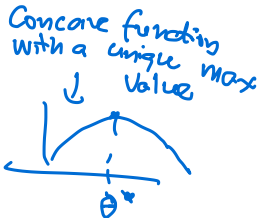
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$$= \arg \max_{\theta} \log P(D | \theta)$$

$$= \arg \max_{\theta} \log (\theta^{n_H} (1 - \theta)^{n_T})$$

$$= \arg \max_{\theta} \underbrace{n_H \log(\theta) + n_T \log(1 - \theta)}_{\text{concave}}$$



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$$0 = \frac{\partial}{\partial \theta} (n_H \log(\theta) + n_T \log(1 - \theta))$$

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$$\begin{aligned}0 &= \frac{\partial}{\partial \theta} (n_H \log(\theta) + n_T \log(1 - \theta)) \\ 0 &= \frac{n_H}{\theta} - \frac{n_T}{1 - \theta}\end{aligned}$$

How to Solve?

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$$\frac{n_H}{\theta} = \frac{n_T}{1-\theta}$$



$$\begin{aligned}0 &= \frac{\partial}{\partial \theta} (n_H \log(\theta) + n_T \log(1 - \theta)) \\ &= \frac{n_H}{\theta} - \frac{n_T}{1 - \theta}\end{aligned}$$

$$(1 - \theta) n_H = n_T \theta$$

$$n_H = \theta (n_H + n_T)$$

$$\theta = (n_H / (n_H + n_T))$$

$$\boxed{\hat{\theta}_{MLE} = \frac{n_H}{n_H + n_T}}$$

Going Back to Our Scenario

- You flip the coin 10 times ...
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$$\hat{\theta}_{MLE} = \frac{n_H}{n_H + n_T} = 0.8 = \frac{8}{8+2}$$

Probability that you get a heads
when you toss the coin is
estimated to be 0.8

Going Back to Our Scenario

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$$\hat{\theta}_{MLE} = \frac{n_H}{n_H + n_T} = \underline{0.8}$$

Here, we are trusting the data completely. But there could be too little data or noisy data

What about Prior Knowledge?

- We believe the coin is *supposed* to be close to 50-50

What about Prior Knowledge?

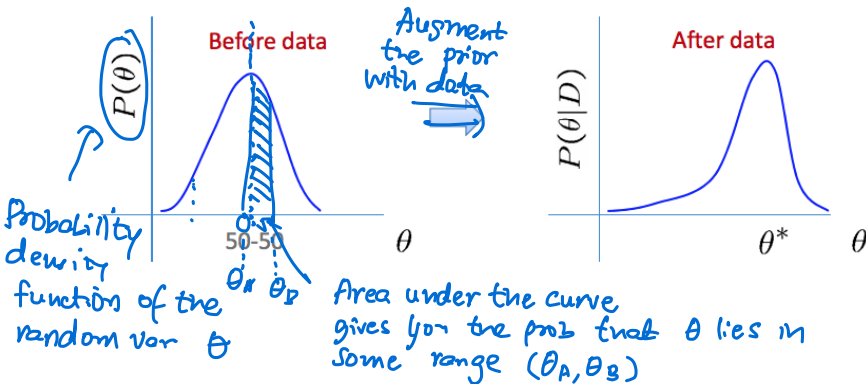
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- How to incorporate prior knowledge?

Bayesian Learning

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- Use Bayes' Rule:

$$P(\theta | D) = \frac{P(D | \theta)P(\theta)}{P(D)}$$

Posterior dist of θ after observing data D →

likelihood of Data →

Prior →

Bayesian Learning

- How to incorporate prior knowledge?
- Use Bayes' Rule:

$$P(\theta | D) = \frac{P(D | \theta)P(\theta)}{P(D)}$$

- Or, equivalently:

$$\underbrace{P(\theta | D)}_{\text{posterior}} \propto \underbrace{P(D | \theta)}_{\text{likelihood}} \underbrace{P(\theta)}_{\text{prior}}$$

We dropped
← $P(D)$ from the
denominator
because it doesn't
depend on θ

Bayesian Learning

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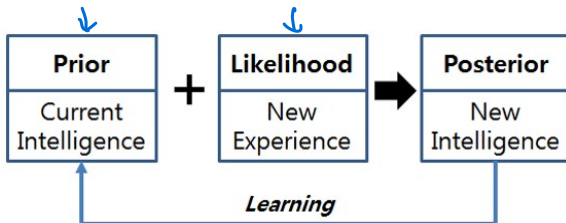
Bayesian Learning

- How to incorporate prior knowledge?
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MAP for Dogecoin

$$\hat{\theta}_{MAP} = \arg \max_{\theta} P(D | \theta) P(\theta)$$

- Recall that $P(D | \theta) = \theta^{n_H} (1 - \theta)^{n_T}$

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- How should we set the prior, $P(\theta)$?
- Common choice is to use the Beta distribution, $\theta \sim \text{Beta}(\alpha, \beta)$:

$$P(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}, \text{ where } B(\alpha, \beta) = \int_0^1 \theta^{\alpha-1} (1-\theta)^{\beta-1} d\theta$$

↑
Normalization const to
ensure that $\int_0^1 P(\theta) d\theta = 1$

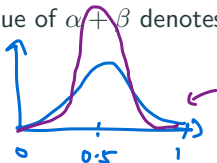
MAP for Dogecoin

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- Recall that $P(D | \theta) = \theta^{n_H} (1 - \theta)^{n_T}$
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- Common choice is to use the **Beta distribution**, $\theta \sim \text{Beta}(\alpha, \beta)$:

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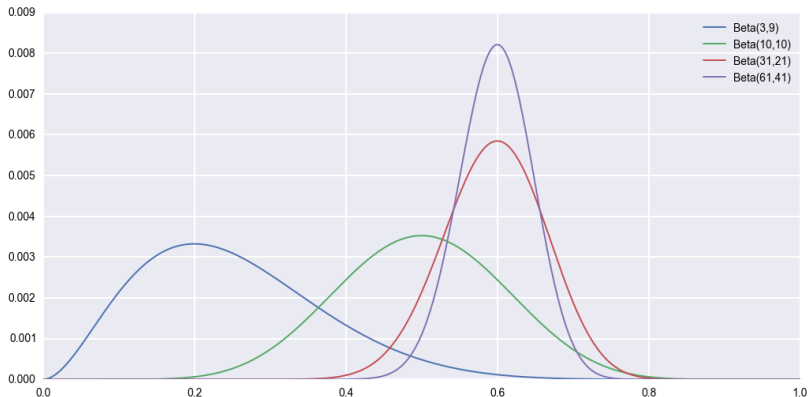
- Interpretation: α = number of expected heads, β = number of expected tails. Larger value of $\alpha + \beta$ denotes more confidence (and smaller variance).



Both priors $P(\theta)$ are centered at 0.5 but the purple one has larger $\alpha + \beta$

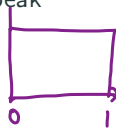
Beta Distribution

← Check Wikipedia article for more special cases



$\frac{\alpha}{\beta}$ controls left/right bias, $\alpha + \beta$ controls height of peak

Special cases $\alpha = \beta = 1 \Rightarrow$ Uniform prior



MAP for Dogecoin

This is because Beta is a
conjugate prior of the Bernoulli & Binomial
distribution
✓

- A benefit of using the *Beta* distribution as a prior is that the posterior will also be a *Beta* distribution:

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Fix $\frac{\alpha}{\beta} = c$
 $\alpha + \beta \rightarrow \infty$
ignore data
 $\hat{\theta}_{map} = \frac{c}{c+1}$

Special case $\leftarrow$

Special case $\rightarrow$ $\alpha = \beta = 1$
 $\frac{n_H}{n_H + n_T} = \text{MLE estimate}$
only trust the data

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$$\begin{aligned}\hat{\theta}_{MAP} &= \arg \max_{\theta} P(D \mid \theta) P(\theta) \\ &= \arg \max_{\theta} \theta^{\alpha + n_H - 1} (1 - \theta)^{\beta + n_T - 1} \\ &= \frac{\alpha + \underline{n_H} - 1}{\alpha + \beta + \underline{n_H} + \underline{n_T} - 2}\end{aligned}$$

- Note that as $\underline{n_H} + \underline{n_T} \rightarrow \infty$, the effect of the prior disappears and we recover the MLE estimate

Putting It All Together

$$\rightarrow \hat{\theta}_{MLE} = \frac{n_H}{n_H + n_T}$$

$$\rightarrow \hat{\theta}_{MAP} = \frac{\alpha + n_H - 1}{\alpha + \beta + n_H + n_T - 2}$$

$\uparrow$

$$\begin{aligned} \lim_{\alpha/\beta \rightarrow \infty} \hat{\theta}_{MAP} &= \lim_{\alpha/\beta \rightarrow \infty} \frac{\alpha}{\alpha + \beta} = \lim_{\alpha/\beta \rightarrow \infty} \frac{\alpha/\beta}{\alpha/\beta + 1} \\ &= \frac{C}{C+1} \end{aligned}$$

Putting It All Together

$$\hat{\theta}_{MLE} = \frac{n_H}{n_H + n_T}$$

$$\hat{\theta}_{MAP} = \frac{\alpha + n_H - 1}{\alpha + \beta + n_H + n_T - 2}$$

True bias,
which we don't
know

- Suppose $\theta^* := 0.5$ and we observe: $D = \{H, H, T, T, T, T\}$

Putting It All Together

$$\hat{\theta}_{MLE} = \frac{n_H}{n_H + n_T} = \frac{2}{2+4}$$
$$\hat{\theta}_{MAP} = \frac{\alpha + n_H - 1}{\alpha + \beta + n_H + n_T - 2}$$

- Suppose $\theta^* := 0.5$ and we observe: $D = \{H, H, T, T, T, T\}$
- Scenario 1: We assume $\theta \sim \text{Beta}(4, 4)$. Which is more accurate – θ_{MLE} or θ_{MAP} ?

$$\theta_{MLE} = \frac{2}{6} = \frac{1}{3} \qquad \theta_{MAP} = \frac{4+2-1}{8+6-2} = \frac{5}{12}$$

Putting It All Together

$$\hat{\theta}_{MLE} = \frac{n_H}{n_H + n_T}$$
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- Scenario 1: We assume $\theta \sim \text{Beta}(4, 4)$. Which is more accurate – θ_{MLE} or θ_{MAP} ?
 - $\theta_{MAP} = 5/12$, $\theta_{MLE} = 1/3$

Putting It All Together

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 - $\theta_{MAP} = 5/12$, $\theta_{MLE} = 1/3$
- Scenario 2: We assume $\theta \sim \text{Beta}(1, 7)$. Which is more accurate – θ_{MLE} or θ_{MAP} ?

Putting It All Together

$$\hat{\theta}_{MLE} = \frac{n_H}{n_H + n_T}$$
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 - $\theta_{MAP} = 5/12$, $\theta_{MLE} = 1/3$
- Scenario 2: We assume $\theta \sim \text{Beta}(1, 7)$. Which is more accurate – θ_{MLE} or θ_{MAP} ?
 - $\theta_{MAP} = 1/6$, $\theta_{MLE} = 1/3$

Bad prior leads to a bad estimate

Why Was This a ML Problem?

Machine learning is: the study of methods that

improve their performance (the accuracy of the predicted probability)

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Machine learning is: the study of methods that

*improve their **performance** (the accuracy of the predicted probability)
on some **task** (predicting the probability of 'heads')*

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Machine learning is: the study of methods that

*improve their **performance*** (the accuracy of the predicted probability)

*on some **task*** (predicting the probability of 'heads')

*with **experience*** (the more coin flips we see, the better our guess)

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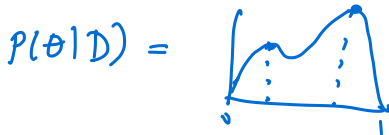
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$$\begin{array}{ccc} \theta_{\text{map}} & \xrightarrow{\alpha=\beta=1} & \theta_{\text{MLE}} \\ & \searrow \frac{\alpha}{\beta}=c & \\ & & \alpha \neq \beta \rightarrow \text{ignore data} \\ & & \frac{\alpha}{\alpha+\beta} \end{array}$$

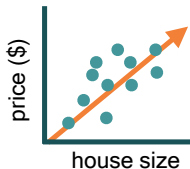
Key idea: these are *choices*. It's important to understand the implications of these choices and evaluate their trade-offs for the problem at hand.



Linear Algebra Review

Recall: Task 1: Regression

How much should you sell your house for?



input: houses & features

learn: $x \rightarrow y$ relationship

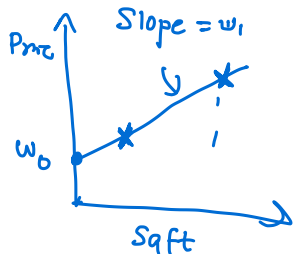
predict: y (continuous)

feature *target*

Handwritten blue annotations: 'feature' with an arrow pointing to 'x' and 'target' with an arrow pointing to 'y' in the 'learn' equation.

Data Can be Compactly Represented by Matrices

- Learn parameters (w_1, w_0) of the orange line $y = w_1x + w_0$



Sq.ft

House 1: 1000 $\times$ w_1 $+ w_0 =$ 200,000

House 2: 2000 $\times$ $w_1 + w_0 =$ 350,000

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- Can represent compactly in matrix notation

$$\underbrace{\begin{bmatrix} 1000 & 1 \\ 2000 & 1 \end{bmatrix}}_X \underbrace{\begin{bmatrix} w_1 \\ w_0 \end{bmatrix}}_w = \underbrace{\begin{bmatrix} 200,000 \\ 350,000 \end{bmatrix}}_y$$

Some Concepts That You Should Know

- Determinants and Matrix Invertibility
- Vector Norms – L1, L2; Inner Products
- Eigenvalues and Eigenvectors
- Singular Value Decomposition
- Covariance Matrices and Positive Semi-definite-ness

Excellent Resources:

- 
- Essence of Linear Algebra YouTube Series by 3Blue1Brown
 - Prof. Gilbert Strang's course at MIT
 - Matrix Cookbook (link posted on the course website)

Matrix Multiplication

- For two matrices $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, their product is:

$$AB = C \in \mathbb{R}^{m \times p} \iff C_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

- Multiplication is undefined with the number of columns in $A \neq$ the number of rows in B (except in case: cA where $c \in \mathbb{R}$ is a scalar)

$${}_m \begin{bmatrix} \overline{A} \\ \hline \end{bmatrix} \begin{bmatrix} \overline{B} \\ \hline \end{bmatrix} = {}_m \begin{bmatrix} \overline{C} \\ \hline \end{bmatrix}$$

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- Special cases:

-  = 0
- • Inner product: $x, y \in \mathbb{R}^n$, $x^T y \in \mathbb{R} = \sum_{i=1}^n x_i y_i$
- Matrix-vector product: $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n \iff Ax \in \mathbb{R}^m$

$$\underline{A} = \begin{bmatrix} | & | & \dots & | \\ a_1 & a_2 & \dots & a_n \\ | & | & \dots & | \end{bmatrix}, \underline{Ax} \in \mathbb{R}^m = \sum_{i=1}^n \underline{a_i x_i}$$

Important Properties

- Associative: $A(\underline{BC}) = \underline{(AB)C}$
- Distributive: $\underline{A}(\underline{B} + \underline{C}) = \underline{AB} + \underline{AC}$
- ***Not*** Commutative: $\underline{AB} \neq \underline{BA}$ (except in special cases)
- Transpose: $\underline{(AB)}^\top = B^\top A^\top$

Matrix Inverse

- The **inverse** of a matrix $A \in \mathbb{R}^{n \times n}$ is a matrix $A^{-1} \in \mathbb{R}^{n \times n}$ such that:

$$AA^{-1} = A^{-1}A = \underline{I_n} = \begin{bmatrix} 1 & 0 & 0 & \dots \\ 0 & 1 & 0 & \dots \\ 0 & 0 & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

↑
Identity
Matrix

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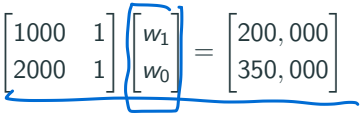
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$$= \frac{1}{-1000} \begin{bmatrix} 1 & -1 \\ -2000 & 1000 \end{bmatrix} \begin{bmatrix} 200,000 \\ 350,000 \end{bmatrix} \quad (3)$$

determinant of matrix $\rightarrow$
 $1000 \times 1 - 2000 \times 1$
 $= -1000$

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$$\begin{bmatrix} w_1 \\ w_0 \end{bmatrix} = \begin{bmatrix} 150 \\ 50,000 \end{bmatrix} \quad (5)$$

Price per sqft
Base price of the piece of land on which the house stands

Norms and Loss Functions

- You could have data from many houses

$$\begin{bmatrix} 1000 & 1 \\ 2000 & 1 \\ 1500 & 1 \\ \vdots & \vdots \\ 2500 & 1 \end{bmatrix}$$

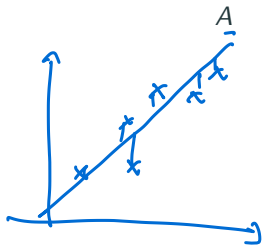
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$\times$

$$\underline{w} =$$

y



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- There isn't a $w = [w_1, w_0]^T$ that will satisfy all equations
- Want to find w that minimizes the **difference** between Aw , y
- But since this a vector, we need an operator that can map the vector $y - Aw$ to a scalar

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ℓ_2 is Euclidean

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- **Question:** What is the ℓ_1 norm of $y - Aw$ for the following problem?

$$\begin{bmatrix} .1 & 1 \\ 2 & 1 \\ 1.5 & 1 \\ 2.5 & 1 \end{bmatrix}$$

A

$\times$

$$\begin{bmatrix} 1.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 2 \\ 3.5 \\ 3 \\ 4.5 \end{bmatrix}$$

$w =$

y

$$Aw = \begin{bmatrix} 2 \\ 2.5 \\ 1.5 + 0.5 \\ 2.5 + 1.5 + 0.5 \end{bmatrix} \quad y = \begin{bmatrix} 2 \\ 3.5 \\ 3 \\ 4.5 \end{bmatrix} \Rightarrow \|y - Aw\|_1 =$$

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- **Answer:** $\|y - Aw\|_1 = \underline{0.5}$

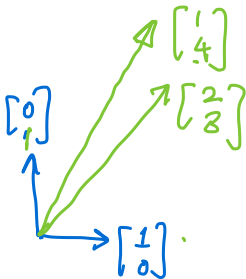
Eigenvalues and Eigenvectors: Matrix as Linear Transformation (see 3Blue1Brown)

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- Now we can express any vector as a linear combination of the above matrix-unit-vector products

$$\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \left(\underline{2} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \underline{1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

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← Eigenvectors
don't change
direction when
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- Viewing A as a linear transformation
 - The vectors that remain unchanged and only get re-scaled are the eigenvectors.

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- Eigenvalues are the roots of $\det(A - \lambda I_n) = 0$
- Eigenvectors are non-zero solutions of $Ax = \lambda x$
- Viewing A as a linear transformation
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- Eigenvalues:

$$\det \left(\begin{bmatrix} 1 - \lambda & 2 \\ 4 & 3 - \lambda \end{bmatrix} \right) = 0$$

$$(1 - \lambda)(3 - \lambda) - 8 = 0$$

$$\lambda^2 - 4\lambda - 5 = 0$$

$$(\lambda - 5)(\lambda + 1) = 0$$

$$\underline{\lambda = 5, \lambda = -1}$$

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Eigenvalue Decomposition

- Group the eigenvectors and eigenvalues into the following matrices.

$$\underline{P} = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$$

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- If the eigenvectors are linearly independent, we can express A as

$$\begin{aligned} \underline{A} &= \underline{P} \underline{\Lambda} \underline{P}^{-1} \\ &= \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}^{-1} \end{aligned}$$

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- V is the matrix of eigenvectors of $A^T A$
 - U is the matrix of eigenvectors of AA^T

What You Should Know

Probability Concepts

- Axioms of probability
- Bayes' Rule
- MLE and MAP procedures
- Expectation and variance of random variables

Linear Algebra Concepts

- Determinants and matrix invertibility
- Vector Norms – ℓ_1 , ℓ_2 ; Inner products
- Eigenvalues and eigenvectors
- Singular Value Decomposition
- Covariance matrices and positive semi-definite-ness